Morning Paper Report - Friday, June 19, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. A recurrent neural network model of prefrontal brain activity during a working memory task

Emilia P Piwek, Mark G Stokes, Christopher Summerfield
2022

Source: Crossref; matched terms: working memory, neural network, neural networks, recurrent neural network

Link: https://doi.org/10.1101/2022.09.02.506349

Goal: Abstract When multiple items are held in short-term memory, cues that retrospectively prioritise one item over another (retro-cues) can facilitate subsequent recall

Method: We extend these findings by analysing the learning dynamics and connectivity patterns of the RNNs, as well as the behaviour of models trained with probabilistic cues, allowing us to make predictions for future studies

Results: We found that the orthogonal-to-parallel geometry transformation observed in the macaque LPFC emerged naturally in RNNs trained to perform the task

Conclusion: Interestingly, the parallel geometry only developed when the cued information was required to be maintained in short-term memory for several cycles before readout, suggesting that it might confer robustness during maintenance

2. Biologically plausible gated recurrent neural networks for working memory and learning-to-learn.

van den Berg AR, Roelfsema PR, Bohte SM
PloS one, 2024

Source: PubMed; matched terms: working memory, neural network, neural networks, recurrent neural network

Link: https://pubmed.ncbi.nlm.nih.gov/39739908/

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

3. Visual field map size constrains working memory precision

Nathan Tardiff, Xingyu Ding, Xiao-Jing Wang, Clayton E Curtis
2026

Source: Crossref; matched terms: working memory, precision, neural network

Link: https://doi.org/10.64898/2026.06.01.729356

Goal: Individual differences in working memory are associated with cognitive functioning and real-world outcomes, including general intelligence, academic achievement, and psychiatric and neurological disorders

Method: To test hypotheses about the potential mechanisms driving our results, we simulated working memory in neural network models with varying numbers of synthetic neurons

Results: In support of our hypothesis, we found that people with larger maps in primary visual cortex and parietal cortex had better memory

Conclusion: In support of our hypothesis, we found that people with larger maps in primary visual cortex and parietal cortex had better memory

4. Effects of Retro-cue Reliability on Visual Working Memory and Attentional Template Efficiency in Visual Search

Alexandre Fortuna, Martin Constant, Dirk Kerzel
2025

Source: Crossref; matched terms: visual working memory, visual search, working memory

Link: https://doi.org/10.31234/osf.io/yzun9_v3

Goal: In the visual search literature, an attentional template refers to a mental representation of target features stored in visual working memory

Method: The dual-state model (Olivers et al., 2011) argues that visual working memory can only be focused on a single representation at a time, and therefore, we can only search for one target at a time

Results: The results suggest that participants allocate their resources based on the reliability of the retro-cue and therefore favor the resource hypothesis

Conclusion: The dual-state model (Olivers et al., 2011) argues that visual working memory can only be focused on a single representation at a time, and therefore, we can only search for one target at a time

5. Effects of Retro-cue Reliability on Visual Working Memory and Attentional Template Efficiency in Visual Search

Alexandre Fortuna, Martin Constant, Dirk Kerzel
2025

Source: Crossref; matched terms: visual working memory, visual search, working memory

Link: https://doi.org/10.31234/osf.io/yzun9_v1

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